

4.8 Centers of Gravity Using Multiple Integrals

Motivation:

Now that we have the notion of double and triple integrals, we can easily get formulas for the center of mass of a lamina which does not have uniform density and center of mass of a three dimensional solid with varying density. We start by looking at the two-dimensional case.

Suppose that a lamina occupies a region D of the xy -plane and its density (in mass per unit area since we view the lamina as two dimensional) at a point in D is given by $\rho(x, y)$ where ρ is a continuous function on D . The

meaning of $\rho(x, y)$ is $\rho(x, y) = \lim \frac{\Delta m}{\Delta A}$ where

- Δm is the mass of the portion of the lamina in a small rectangle containing (x, y) ,
- ΔA is the area of the small rectangle,
- and the limit is taken as the dimensions of the rectangle approach 0.

Based upon how $\rho(x, y)$ is defined, it makes sense intuitively that the mass of the lamina is given by

$$m = \iint_D \rho(x, y) dA .$$

Definition: The coordinates $(\bar{x}, \bar{y})$ of the center of mass of a lamina occupying the region D and having density function $\rho(x, y)$ are

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA \quad \text{and} \quad \bar{y} = \frac{1}{m} \iint_D y \rho(x, y) dA$$

Note: Three double integrals are required to find the center of mass of a lamina.

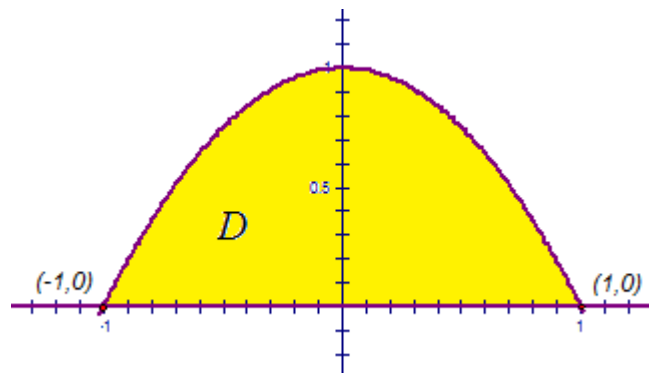
Note: Intuitively, one can view the double integrals for each coordinate of the center of mass as “weighted averages”.

Note: One will often hear $M_x = \iint_D y \rho(x, y) dA$ called the moment of the lamina about the x -axis, and will hear

$M_y = \iint_D x \rho(x, y) dA$ called the moment of the lamina about the y -axis.

Example 1: Suppose that a lamina occupies the region $D = \{(x, y) \mid y \leq 1 - x^2, y \geq 0\}$ and has density function $\rho(x, y) = y$. Find the center of mass of the lamina.

A picture of the region D is:



So, we note that the mass of the lamina is

$$m = \iint_D y dA = \int_{-1}^1 \left[\int_0^{1-x^2} y dy \right] dx = \int_{-1}^1 \left[\frac{1}{2} y^2 \right]_{y=0}^{y=1-x^2} dx = \frac{1}{2} \int_{-1}^1 (1 - 2x^2 + x^4) dx = \frac{1}{2} \left(x - \frac{2}{3} x^3 + \frac{1}{5} x^5 \right) \Big|_{-1}^1 = 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}.$$

Then, by definition,

$$\bar{x} = \frac{15}{8} \int_{-1}^1 \left(\int_0^{1-x^2} xy dy \right) dx = \frac{15}{8} \int_{-1}^1 \left(\frac{xy^2}{2} \right)_{y=0}^{y=1-x^2} dx = \frac{15}{16} \int_{-1}^1 (x - 2x^3 + x^5) dx = \frac{15}{16} \left(\frac{1}{2} x^2 - \frac{1}{2} x^4 + \frac{1}{6} x^6 \right) \Big|_{-1}^1 = 0.$$

Then, by definition,

$$\begin{aligned} \bar{y} &= \frac{15}{8} \int_{-1}^1 \left(\int_0^{1-x^2} y^2 dy \right) dx = \frac{15}{8} \int_{-1}^1 \left(\frac{y^3}{3} \right)_{y=0}^{y=1-x^2} dx = \frac{15}{24} \int_{-1}^1 (1 - x^2)^3 dx = \frac{5}{8} \int_{-1}^1 (1 - 3x^2 + 3x^4 - x^6) dx \\ &= \frac{5}{8} \left(x - x^3 + \frac{3}{5} x^5 - \frac{1}{7} x^7 \right) \Big|_{-1}^1 = \frac{5}{4} \left(1 - 1 + \frac{3}{5} - \frac{1}{7} \right) = \frac{4}{7}. \end{aligned}$$

Thus, the center of mass of the lamina is $\left(0, \frac{4}{7} \right)$.

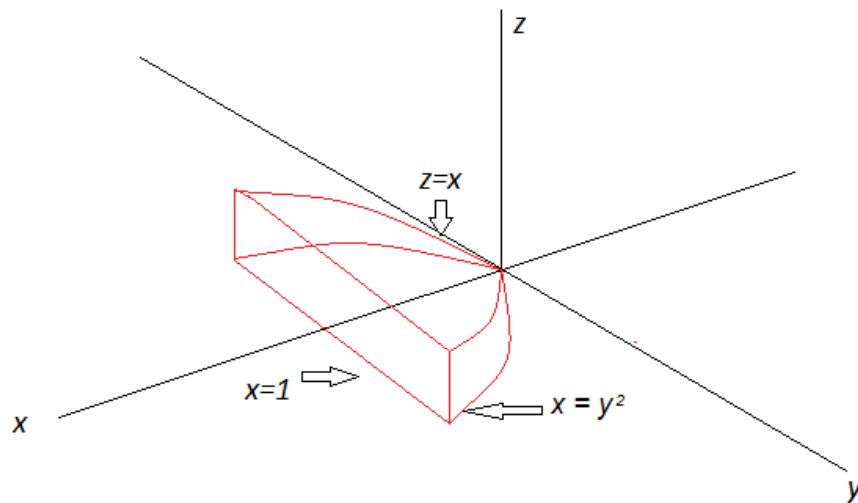
We now explore how to find the center of mass of a three dimensional solid with varying density.

Definition: Suppose that the density function of a solid object that occupies the region E is the continuous function $\rho(x, y, z)$ (the unit for this function is mass per unit volume). The center of mass of this object is

$$(\bar{x}, \bar{y}, \bar{z}) \text{ where } \bar{x} = \frac{1}{m} \iiint_E x \rho(x, y, z) dV, \bar{y} = \frac{1}{m} \iiint_E y \rho(x, y, z) dV, \bar{z} = \frac{1}{m} \iiint_E z \rho(x, y, z) dV, \text{ and } m = \iiint_E \rho(x, y, z) dV.$$

Example 2: Find the center of mass of the solid that occupies the region bounded by the graphs of $x = y^2$, $x = z$, $z = 0$, and $x = 1$ which has density function $\rho(x, y, z) = 1$.

We will call the solid E . A picture of the solid is:



So, we have that $E = \{(x, y, z) \mid x \geq y^2, x \geq z, z \geq 0, x \leq 1\}$. From these inequalities and the above picture, we note that $0 \leq z \leq x$ and the projection of E onto the xy -plane is $D = \{(x, y) \mid y^2 \leq x \leq 1, -1 \leq y \leq 1\}$. So, we have that (the integration computation is omitted):

$$\begin{aligned} m &= \int_{-1}^1 \int_{y^2}^1 \int_0^x 1 dz dx dy = \frac{4}{5} & \bar{x} &= \frac{1}{m} \int_{-1}^1 \int_{y^2}^1 \int_0^x x dz dx dy = \frac{5}{7} \\ \bar{y} &= \frac{1}{m} \int_{-1}^1 \int_{y^2}^1 \int_0^x y dz dx dy = 0 & \bar{z} &= \frac{1}{m} \int_{-1}^1 \int_{y^2}^1 \int_0^x z dz dx dy = \frac{5}{14} \end{aligned}$$

Thus, the answer is that the center of mass of the solid is $\left(\frac{5}{7}, 0, \frac{5}{14}\right)$.